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# Generating Physically Consistent Molecules with Energy-Based Models

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## Abstract

Molecules in equilibrium follow a Boltzmann distribution, making the underlying energy landscape a physically grounded modeling objective. However, such landscapes are difficult to learn from data and, once learned, hard to sample from. Diffusion and flow-matching models sidestep these difficulties by learning a time-conditional score or transport field between noise and data, losing the energy inductive bias in exchange for a more tractable training objective. We introduce EBMol, an energy-based model (EBM) that restores this inductive bias by learning an atom-additive scalar potential without explicit simulation during training. Our method employs a flow-inspired Restoring Field Matching objective to approximate the energy landscape. We adopt the Mirror-Langevin algorithm for sampling, enabling unified updates of atomic positions and types, and incorporate parallel tempering for inference-time compute scaling. EBMol is the first EBM for 3D molecular generation to achieve state-of-the-art performance on QM9 and GEOM-Drugs. Moreover, we show that the learned energy landscape serves as a principled quality metric for ranking and filtering configurations, and demonstrate controllable generation without retraining through shape-steered sampling via potential composition and zero-shot linker design.\*

## 1 Introduction

Molecular states at thermal equilibrium are governed by the Boltzmann distribution  $p(\mathbf{x}) \propto \exp(-E(\mathbf{x})/(k_B T))$ , in which the probability of a configuration is determined entirely by its potential energy. The standard datasets for molecular generation, QM9 [1] and GEOM [2], are not drawn from this distribution. Instead, they contain geometry-optimized structures, each relaxed to a local energy minimum by density functional theory or semi-empirical methods. What is physically meaningful in these datasets is where the energy minima lie. The relative frequencies at which different molecular structures appear reflect enumeration strategies and computational budgets, not thermodynamic weights [3]. A scalar energy landscape whose minima coincide with stable configurations is therefore the natural modeling target: it captures exactly the physical content of the data and takes the functional form of a Boltzmann distribution,  $p_\theta(\mathbf{x}) \propto \exp(-E_\theta(\mathbf{x})/\tau)$ .

Learning and sampling from such landscapes, however, have remained open problems. Training a physically-calibrated energy function requires Boltzmann-distributed data [4] such as SPICE [5]

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\*Code available at <https://github.com/griesbchr/EBMol>.

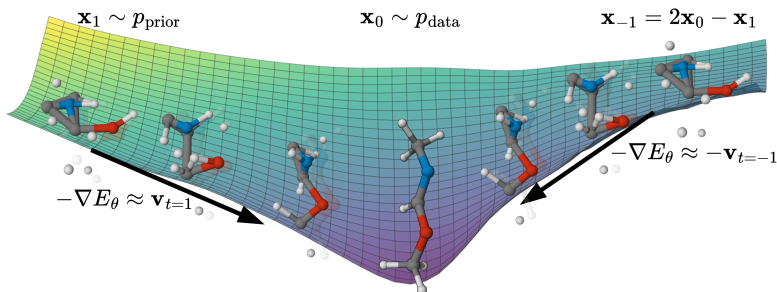


Figure 1: Overview of Restoring Field Matching. The RFM loss shapes a scalar energy landscape with data points  $x_0 \sim p_{\text{data}}$  as local minima by training the negative gradient  $-\nabla E_\theta$  to point toward data from both sides: from the prior (interpolation,  $t \in [0, 1]$ ) and from beyond the data (extrapolation,  $t \in [-1, 0]$ ).

or MD17 [6], but datasets with this property target interatomic interactions and conformational sampling rather than chemical diversity and are unsuitable for generation. Sampling is equally difficult: generating molecules from an energy function typically relies on Markov Chain Monte Carlo (MCMC) methods, which must traverse the extremely rugged, unphysical regions between noise and valid structures [7]. Diffusion [8, 9] and flow matching models [10, 11] sidestep both barriers by directly learning noise-to-data transport maps through time-conditional vector fields, achieving state-of-the-art results. However, the resulting models provide no explicit energy function and no mechanism to evaluate, rank, or compose configurations at inference time.

In this work, we introduce *EBMol*, an Energy-Based Model (EBM) for Molecules that addresses both barriers. For training, we propose Restoring Field Matching (RFM, Figure 1), a flow-matching-inspired objective that shapes the energy landscape with data points as local minima, without requiring simulation or Boltzmann-distributed data. For sampling, we adapt the Mirror Langevin Algorithm (MLA) [12] to molecular state spaces, applying Euclidean updates on coordinates and simplex-constrained updates on atom types jointly within each step. Given that the learned energy is an explicit, persistent scalar function rather than an implicit mapping from noise to data, capabilities that time-conditional models cannot naturally provide follow directly: energy-based ranking and filtering of generated samples, composable generation through potential composition, and zero-shot conditional generation by fixing molecular fragments.

Our contributions are as follows:

- We introduce Restoring Field Matching (RFM), a simulation-free training objective for tractable learning of per-atom additive energy landscapes, and establish theoretical guarantees for the resulting optimization problem.
- We adapt the Mirror Langevin Algorithm to the mixed continuous-simplex state space of molecules, unifying coordinate and atom types updates in a single framework, combined with parallel tempering enabling inference-time compute scaling.
- EBMol is the first EBM for 3D molecular generation to achieve state-of-the-art performance on QM9 and GEOM-Drugs. This demonstrates that the benefits of energy-based modeling, including physically meaningful scoring, inference-time ranking, composable shape steering, and zero-shot linker design, are attainable without sacrificing generation quality.

## 2 Related work

**Energy-based modeling and related time-unconditional frameworks.** Energy-based models [13] have traditionally been trained via MCMC-based maximum likelihood [14] or score matching [15], both of which sidestep the partition function but at substantial cost: MCMC is expensive and prone to mode collapse [16], while score matching either requires Hessian traces or re-introduces a time variable through noise conditioning [15]. Despite the recent popularity of time-conditional diffusion [8, 9] and flow matching methods [10, 11], several recent works have revisited time-

unconditional EBMs. Energy Matching [17] learns an energy landscape in two stages, combining transport and contrastive-divergence. Equilibrium Matching [18] is conceptually closest to RFM, also enforcing stationarity at data by scaling the target field to zero, but learns an implicit non-conservative vector field rather than an explicit scalar energy. Also, neither method addresses discrete-continuous state spaces. Within the molecular domain specifically, time-unconditional EBM training has previously been attempted only on highly restricted settings by [19], who train an equivariant EBM via simulation-based contrastive divergence on a single-stoichiometry subset of QM9, thus neglecting variable molecule size, mixed discrete-continuous state, or modern benchmarks.

**Generative modeling of 3D molecules.** Various different generative modeling paradigms have been explored for 3D molecular generation. Besides autoregressive models [20–22], field-based approaches [23, 24], and, most prominently, diffusion and flow matching models. These methods differ primarily in their treatment of bonds: bond-explicit methods use dedicated supervision [25–30], while bond-implicit approaches, which include our work, rely on emergent geometric accuracy to recover connectivity post-hoc via OpenBabel [31, 26] or predefined lookup tables [32]. The seminal EDM [32] introduced equivariant diffusion via the EGNN [33] backbone to molecular modeling. Subsequent work refined this line through latent-space variants [34], alternative architectures and training [35], flow matching [36–38], and Bayesian flow networks [39]. Several works that incorporate energy-inspired components into their generative pipeline are related to our approach. These include using Amber and MACE force fields [40, 7], interpreting self-consistency residuals as energies in iteratively refined flow models [41], or employing energy-driven objectives for conformer generation [42]. Crucially, however, none of these methods learn a persistent, scalar energy landscape that drives end-to-end sampling.

**Mirror Langevin algorithm and constrained sampling.** The Mirror Langevin Algorithm [12] extends mirror descent [43, 44], originally developed for constrained optimization, to sampling on constrained domains. Theoretical analyses on convex domains have established convergence rates [45, 46] and Metropolis-adjusted extensions [47]. While MLA has recently been adapted for generative modeling on homogeneous constrained domains [48], EBMol is, to the best of our knowledge, the first unified mixed-geometry Langevin framework for joint continuous-discrete deep generative modeling.

### 3 Method

We propose *EBMol*, a time-unconditional energy-based model that learns a scalar potential  $E_\theta(\mathbf{x})$  over molecular configurations. EBMol combines two largely independent components: Restoring Field Matching (RFM) for tractable shaping of the energy landscape, and the Mirror Langevin Algorithm (MLA) [12] for sampling from it.

#### 3.1 Problem formulation

A molecule with  $N$  atoms is represented as a joint state  $\mathbf{x} = (\mathbf{c}, \mathbf{p})$ , where  $\mathbf{c} \in \mathbb{R}^{N \times 3}$  are atomic coordinates in Euclidean space and  $\mathbf{p} = (\mathbf{p}_0, \dots, \mathbf{p}_{N-1})$  are per-atom categorical distributions with  $\mathbf{p}_i \in \Delta^{K-1} = \{\mathbf{p}_i \in \mathbb{R}^K \mid p_{i,k} \geq 0, \sum_k p_{i,k} = 1\}$  over  $K$  chemical elements. Bonds are not explicitly modeled; instead, connectivity and bond orders are inferred post-hoc from  $(\mathbf{c}, \mathbf{p})$ , placing the burden of chemical validity primarily on geometric consistency. We parameterize a time-unconditional energy  $E_\theta(\mathbf{c}, \mathbf{p})$  over this mixed state space. To handle variable-sized molecules, we impose a permutation-invariant additive decomposition over atoms  $E_\theta(\mathbf{c}, \mathbf{p}) = \sum_{i=1}^N E_\theta^{(i)}(\mathbf{c}, \mathbf{p})$ .

#### 3.2 Learning with the Restoring Field Matching objective

Learning a meaningful energy landscape is challenging. Maximum-likelihood training relies on MCMC [14], and score matching either requires Hessian traces [49] or introduces a time variable [8, 9]. We instead construct the landscape *geometrically*: using flow-matching-style interpolation, we regress the energy gradient toward a field that points back to the data from both sides of the manifold, making data points local minima.

**Constructing the restoring field.** Standard flow matching defines an interpolant  $\mathbf{x}_t$  between  $\mathbf{x}_1 \sim p_{\text{prior}}$  at  $t = 1$  and  $\mathbf{x}_0 \sim p_{\text{data}}$  at  $t = 0$ , with a target velocity that is nonzero everywhere on the interval. Data is thus a trajectory endpoint, not a stationary point of any field. We begin by extending the interpolation to  $t \in [-1, 1]$ , so the interpolant  $\mathbf{x}_t$  passes *through* the data rather than terminating at it. We construct this extended interpolant separately for the two components of  $\mathbf{x} = (\mathbf{c}, \mathbf{p})$ . For continuous coordinates, we first align  $\mathbf{c}_1$  to  $\mathbf{c}_0$  using the equivariant optimal-transport coupling to reduce trajectory curvature. The aligned noise is denoted  $\mathbf{c}_1$ , and we define a linear trajectory  $\mathbf{c}_t = \mathbf{c}_0 + t(\mathbf{c}_1 - \mathbf{c}_0)$ ,  $t \in [-1, 1]$ . For categorical features, linear extrapolation leaves the simplex. We instead use a reflection,  $\mathbf{p}_t = \mathbf{p}_0 + |t|(\mathbf{p}_1 - \mathbf{p}_0)$ , which remains in  $\Delta^{K-1}$  and is symmetric around  $t = 0$ .

Along each interpolant, we define a target field that points toward the data  $\mathbf{x}_0$ . Building on the natural choice  $\text{sign}(t)(\mathbf{x}_0 - \mathbf{x}_1)$ , we directly encode stationarity at the data by scaling the target with a smoothing function  $b : [0, 1] \rightarrow [0, 1]$  satisfying  $b(0) = 0$  and  $b(1) = 1$ , so that the target vanishes at  $t = 0$ :

$$\mathbf{u}^{\mathbf{c}} = b(|t|) \cdot \text{sign}(t) \cdot (\mathbf{c}_0 - \mathbf{c}_1), \quad \mathbf{u}^{\mathbf{p}} = b(|t|) \cdot (\mathbf{p}_0 - \mathbf{p}_1). \quad (1)$$

The model is trained to match  $-\nabla_{\mathbf{x}} E_{\theta}(\mathbf{x}_t)$  against this target, as described below.

**Training objective.** Denoting  $\mathbf{v} := -\nabla_{\mathbf{x}} E_{\theta}(\mathbf{x}_t)$  and  $\mathbf{u} := (\mathbf{u}^{\mathbf{c}}, \mathbf{u}^{\mathbf{p}})$ , the RFM objective is given by

$$\mathcal{L}_{\text{RFM}} = \mathbb{E}_{\mathbf{x}_0, \mathbf{x}_1, t \sim U[-1, 1]} \left[ \|\mathbf{v} - \mathbf{u}\|^2 \right]. \quad (2)$$

While the restoring target specifies the energy *gradient*, it leaves the energy *value* at the data unconstrained. Without regularization, different data points may sit at different absolute energies, biasing the sampler toward artificially deeper basins. We therefore add a per-atom regularization term that anchors clean-data energies toward zero:

$$\mathcal{L}_{\text{reg}} = \mathbb{E}_{\mathbf{x}_0 \sim p_{\text{data}}} \left[ \frac{1}{N} \sum_{i=1}^N E_{\theta}^{(i)}(\mathbf{x}_0)^2 \right]. \quad (3)$$

The full training loss is then given by

$$\mathcal{L} = \mathcal{L}_{\text{RFM}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}, \quad (4)$$

where  $\lambda_{\text{reg}}$  controls the regularization strength. Training is summarized in Algorithm 1.

### 3.3 Sampling with annealed Mirror-Langevin algorithm

We sample from  $p_{\theta}(\mathbf{x}) \propto \exp(-E_{\theta}(\mathbf{x})/\tau)$  via Langevin dynamics. While standard Langevin suffices for coordinates, it pushes atom types outside the simplex. The Mirror Langevin Algorithm (MLA) [12] resolves this by performing updates in an unconstrained dual space and projecting back through a mirror map.

Let  $\Phi : \mathcal{D} \rightarrow \mathbb{R}$  be a strictly convex, twice-differentiable mirror potential on the constrained domain  $\mathcal{D}$  such that its gradient  $\nabla \Phi$  maps  $\mathcal{D}$  to an unconstrained dual space, and the inverse map  $(\nabla \Phi)^{-1}$  projects back to  $\mathcal{D}$ . The MLA update with step size  $\eta$ , temperature  $\tau$ , and noise  $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  reads

$$\mathbf{y}^{(k+1)} = \nabla \Phi(\mathbf{x}^{(k)}) - \eta \nabla_{\mathbf{x}} E_{\theta}(\mathbf{x}^{(k)}) + \sqrt{2\eta\tau} (\nabla^2 \Phi(\mathbf{x}^{(k)}))^{1/2} \boldsymbol{\xi}, \quad (5)$$

$$\mathbf{x}^{(k+1)} = (\nabla \Phi)^{-1}(\mathbf{y}^{(k+1)}). \quad (6)$$

The factor  $(\nabla^2 \Phi)^{1/2}$  is the Riemannian metric correction induced by the mirror potential, ensuring the update is consistent with the underlying mirror Langevin SDE.

**Continuous coordinates and categorical features.** The choice of mirror potential  $\Phi$  specializes MLA to each component of  $\mathbf{x} = (\mathbf{c}, \mathbf{p})$ . For  $\mathbf{c} \in \mathbb{R}^{N \times 3}$ , the Euclidean potential  $\Phi^{\mathbf{c}}(\mathbf{c}) = \frac{1}{2} \|\mathbf{c}\|^2$  reduces MLA to the Unadjusted Langevin Algorithm. For atom types  $\mathbf{p} \in \Delta^{K-1}$ , the negative entropy  $\Phi^{\mathbf{p}}(\mathbf{p}) = \sum_i p_i \log p_i$  maps the simplex to logit space via  $\nabla \Phi^{\mathbf{p}} = \log \mathbf{p} + \mathbf{1}$ , with  $(\nabla \Phi^{\mathbf{p}})^{-1} = \text{softmax}$  projecting back exactly. Since the negative-entropy mirror map does not satisfy the self-concordance condition of [12] near the simplex boundary, we clamp  $\mathbf{p} \geq \epsilon$  before each update (see Appendix E for details). The full update is given in Algorithm 2.

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**Algorithm 1** RFM training step

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**Require:** Energy  $E_\theta$ , smoothing function  $b(t)$ 

- 1: Sample  $\mathbf{x}_0 = (\mathbf{c}_0, \mathbf{p}_0) \sim p_{\text{data}}$
  - 2: Sample  $\mathbf{x}_1 = (\mathbf{c}_1, \mathbf{p}_1) \sim p_{\text{prior}}$
  - 3: Sample  $t \sim \mathcal{U}(-1, 1)$
  - 4:  $T_{\text{OT}} \leftarrow \text{OTSolver}(\mathbf{c}_1, \mathbf{c}_0)$
  - 5:  $\mathbf{c}_t \leftarrow \mathbf{c}_0 + t(T_{\text{OT}} \mathbf{c}_1 - \mathbf{c}_0)$
  - 6:  $\mathbf{p}_t \leftarrow \mathbf{p}_0 + |t|(\mathbf{p}_1 - \mathbf{p}_0)$
  - 7:  $\mathbf{v} \leftarrow -\nabla_{\mathbf{x}} E_\theta(\mathbf{c}_t, \mathbf{p}_t)$
  - 8:  $\mathbf{u}_{\mathbf{c}} \leftarrow b(|t|) \text{sign}(t) (\mathbf{c}_0 - \mathbf{c}_1)$
  - 9:  $\mathbf{u}_{\mathbf{p}} \leftarrow b(|t|) (\mathbf{p}_0 - \mathbf{p}_1)$
  - 10:  $\mathcal{L}_{\text{reg}} \leftarrow \frac{\lambda_{\text{reg}}}{N} \sum_{i=1}^N E_\theta^{(i)}(\mathbf{x}_0)^2$
  - 11:  $\mathcal{L} \leftarrow \|\mathbf{v} - \mathbf{u}\|^2 + \mathcal{L}_{\text{reg}}$
  - 12: **Update**  $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$
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**Algorithm 2** Mirror-Langevin sampling step

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**Require:** State  $(\mathbf{c}, \mathbf{p})$ ,  $E_\theta$ , step size  $\eta$ , temperature  $\tau$ , noise scales  $\sigma_{\mathbf{c}}, \sigma_{\mathbf{p}}$ , clipping  $\epsilon$ 

- 1:  $\mathbf{g}_{\mathbf{c}} \leftarrow \nabla_{\mathbf{c}} E_\theta(\mathbf{c}, \mathbf{p}), \quad \mathbf{g}_{\mathbf{p}} \leftarrow \nabla_{\mathbf{p}} E_\theta(\mathbf{c}, \mathbf{p})$
  - Continuous update:**
  - 2:  $\xi_{\mathbf{c}} \sim \mathcal{N}(\mathbf{0}, \sigma_{\mathbf{c}}^2 \mathbf{I})$
  - 3:  $\mathbf{c}' \leftarrow \mathbf{c} - \eta \mathbf{g}_{\mathbf{c}} + \sqrt{2\eta\tau} \xi_{\mathbf{c}}$
  - 4:  $\mathbf{c}' \leftarrow \mathbf{c}' - \frac{1}{N} \sum_i \mathbf{c}'_i$
  - Categorical update:**
  - 5:  $\tilde{\mathbf{p}} \leftarrow \max(\mathbf{p}, \epsilon)$
  - 6:  $\xi_{\mathbf{p}} \sim \mathcal{N}(\mathbf{0}, \sigma_{\mathbf{p}}^2 \mathbf{I})$
  - 7:  $\mathbf{y} \leftarrow \log \tilde{\mathbf{p}} - \eta \mathbf{g}_{\mathbf{p}} + \sqrt{2\eta\tau} \tilde{\mathbf{p}}^{-1/2} \odot \xi_{\mathbf{p}}$
  - 8:  $\mathbf{p}' \leftarrow \text{Softmax}(\mathbf{y})$
  - 9: **return**  $(\mathbf{c}', \mathbf{p}')$
- 

### 3.4 Parallel tempering

We run  $M$  parallel chains at a fixed ladder of temperatures  $\tau_1 > \tau_2 > \dots > \tau_M$ , with periodic Metropolis-Hastings swaps between adjacent levels according to energies  $E$  with acceptance probability

$$\min\left(1, \exp\left[\left(E(\mathbf{x}_j) - E(\mathbf{x}_i)\right)\left(\frac{1}{\tau_i} - \frac{1}{\tau_j}\right)\right]\right). \quad (7)$$

Hot chains explore broadly, cold chains refine, and swaps propagate discoveries down the ladder. Periodically, we harvest samples from the coldest chain and replace them with prior samples at the same coldest level, which are swapped upward by the energy-based exchange mechanism. Subsequently, harvested samples undergo a short  $\tau = 0$  relaxation. Parallel tempering offers a built-in mechanism for inference-time compute scaling by increasing the number of tempering rounds between harvests. Additional swaps and exploration before samples are extracted yields progressively higher-quality molecules at the cost of longer sampling.

### 3.5 Analysis

We establish two theoretical properties of the RFM objective that characterize the minimization problem defined by (4). The analysis considers  $E$  as a general element of a function space rather than a specific network architecture. For sufficiently expressive networks, these properties are expected to hold approximately in practice, which our experiments in Section 4 support. Formal assumptions and complete proofs for both properties are given in Appendix A.

The first result ensures that the training problem is well-posed.

**Proposition 1** (Existence of solutions, informal). *Under appropriate assumptions on  $p_{\text{prior}}$  and  $p_{\text{data}}$ , for every  $\lambda_{\text{reg}}$  the following problem admits a unique solution:*

$$\min_E \mathcal{L}(E).$$

Given existence, we characterize the solution’s behavior at the two extremes of the regularization parameter.

**Proposition 2** (Convergence for small and large regularization parameters, informal). *Let  $E^*(\mathbf{x}, \lambda)$  be the minimizer of  $\mathcal{L}$  for  $\lambda_{\text{reg}} = \lambda$ . The following holds true:*

- (i)  $\underline{\lambda = \infty}$ : *For every sequence  $\lambda_n \rightarrow \infty$ , there exists a subsequence  $(\lambda_{n_k})_k$  such that  $E^*(\cdot, \lambda_{n_k}) \rightarrow 0$   $p_0$ -a.e.; figuratively:  $E^*(\mathbf{x}_0, \lambda) \approx 0$  for  $\lambda \gg 0$  and  $\mathbf{x}_0 \sim p_{\text{data}}$ .*
- (ii)  $\underline{\lambda = 0}$ : *For every sequence  $\lambda_n \rightarrow 0$ , there exists a subsequence  $(\lambda_{n_k})_k$  such that  $\nabla E^*(\cdot, \lambda_{n_k})$  converges to the projection of the prescribed restoring field  $\mathbf{u}$  onto the space of all conservative vector fields. That is,  $\nabla E^*(\cdot, \lambda_{n_k})$  converges to the minimizer of the functional*

$$E \mapsto \mathbb{E}_{\mathbf{x}_0, \mathbf{x}_1, t \sim \mathcal{U}[-1, 1]} [\|\nabla E(\mathbf{x}_t) - \mathbf{u}\|^2] \quad (8)$$

Moreover, if  $\mathbf{x} \mapsto \mathbb{E}[\mathbf{u}|x_t = x]$  is conservative

$$\lim_{k \rightarrow \infty} \nabla E^*(\mathbf{x}, \lambda_{n_k}) = -\mathbb{E}[\mathbf{u}|x_t = x] \quad p_x\text{-a.e.} \quad (9)$$

The above results confirm our modeling approach theoretically: strong regularization anchors the energy to zero at the data, while the gradient matching term leads to the L2 projection of the target restoring field onto the space of conservative vector fields. At any finite  $\lambda_{\text{reg}}$ , the minimizer interpolates between these two extremes. However, the central modeling claim, that data points are local minima of  $E_\theta$ , is not established by the propositions above. It rests on the target construction ( $b(0) = 0$  encodes zero gradient at  $t = 0$ ) and is verified empirically in Appendix B.

## 4 Experiments

### 4.1 Setup

**Datasets.** We evaluate EBMol on the two standard benchmarks for unconditional 3D molecular generation, QM9 [1] and GEOM-Drugs [2]. QM9 contains 134k small organic molecules with up to nine heavy atoms across five element types (H, C, N, O, F), exhaustively enumerating all stable molecules within this space at DFT level. GEOM-Drugs poses a more challenging benchmark, comprising roughly 430k drug-like molecules with an average size of 44 atoms (up to 181) across 15 element types. This dataset is substantially more diverse than QM9.

**Implementation.** To ensure SE(3) invariance of the energy, we employ a standard EGNN [33] network, propagating equivariant messages across a fully connected atomic graph. We compute the per-atom energies by applying a two-layer MLP to the invariant node features of the final layer, omitting the final equivariant feature update. The total molecule energy is calculated by summing over the node energies, preserving SE(3) invariance of  $E_\theta$ . During training, we use a diagonal anisotropic Gaussian position prior whose per-molecule covariance is set to the principal-component eigenvalues of a training molecule with the same atom count, introducing a mild data-dependent shape bias that improves training stability and performance. For atom types, we use a Dirichlet distribution with concentration  $1/K$ , where  $K$  is the number of atom types. We employ  $\tanh(\cdot)$  as smoothing function. Pseudocode for the training and sampling algorithms is presented in Algorithms 1 and 2. During generation, we sample the atom count  $N$  from the empirical training distribution and base the parallel tempering swaps on the largest per-atom energy  $\max_i E_\theta^{(i)}(\mathbf{x})$ . A full list of hyperparameters, more experiment details and ablations on design choices can be found in Appendices D and E.

**Baselines and evaluation.** We compare against a variety of established bond-implicit baselines for molecular generation [32, 40, 36, 50, 34, 38, 39, 35]. All baselines share the same EGNN network architecture [33], with the exception of SLDM, employing a modified EGNN variant based on UniGEM [51] with more parameters (2.2M vs 4.2M for GEOM-Drugs). On GEOM-Drugs, we restrict baselines to methods with publicly available checkpoints to ensure consistent re-evaluation under our revised evaluation protocol.

On QM9 we follow the standard protocol of [32], reporting atom and molecule stability, validity, uniqueness, and novelty. On GEOM-Drugs, recent models have begun to saturate this protocol: its QM9-calibrated bond and valency lookup tables yield a data baseline of only 86.5% atom stability, a ceiling already exceeded by SLDM [35]. To restore this benchmark’s discriminative power, we adopt a revised protocol. We perform bond inference via OpenBabel [31], which additionally infers formal charges and aromaticity. This additional information unlocks the usage of the valency tables proposed by [52], which are specifically calibrated for the element and bond-type diversity encountered in GEOM-Drugs. Furthermore, we use the OpenBabel-inferred molecules for RDKit sanitization with stereochemistry assignment for more expressive validity, uniqueness, and novelty metrics. Under this revised protocol, the data baseline rises to 99.6% atom stability (from 86.5%) and 88.2% molecule stability (previously  $\sim 0\%$ ), indicating a much more faithful assessment of chemical validity. We additionally report the Vendi score [53] for molecular diversity and the physical metric suite of [52], which assesses differences in bond lengths, bond angles and torsion angles between generated molecules and their GFN2-xTB-optimized [54] counterparts, alongside the associated mean and median relaxation energies.

## 4.2 Unconditional generation

Table 1: Standard molecular generation metrics on QM9, computed on 10k generated samples per model. Results marked with \* are from our own experiments. Unless otherwise noted in the subscript, models use 1000 NFEs.

| Method                 | Atom $\uparrow$<br>Stability | Molecule $\uparrow$<br>Stability | Validity $\uparrow$ | Uniqueness $\uparrow$ | Valid &<br>Unique $\uparrow$ | Novelty $\uparrow$ |
|------------------------|------------------------------|----------------------------------|---------------------|-----------------------|------------------------------|--------------------|
| Data                   | 99.40                        | 95.30                            | 97.70               | 99.97                 | 97.60                        | –                  |
| EDM                    | 98.70                        | 82.00                            | 91.90               | –                     | 90.70                        | 65.70              |
| EDM-Bridge             | 98.80                        | 84.60                            | 92.00               | –                     | 90.70                        | –                  |
| EquiFM <sub>210</sub>  | 98.90                        | 88.30                            | 94.70               | –                     | 93.50                        | –                  |
| END                    | 98.90                        | 89.10                            | 94.80               | –                     | 92.60                        | –                  |
| GEOLDM                 | 98.90                        | 89.40                            | 93.80               | –                     | 92.70                        | 57.00*             |
| GOAT                   | 99.20                        | –                                | 92.90               | 99.00                 | –                            | <b>78.60</b>       |
| GeoBFN <sub>2000</sub> | 99.31                        | 93.32                            | 96.88               | –                     | <b>92.41</b>                 | 65.30              |
| SLDM                   | 99.43                        | 95.42                            | 97.07               | –                     | 90.42                        | –                  |
| SLDM*                  | 99.07                        | 92.32                            | 95.53               | 95.28                 | 91.02                        | 42.46              |
| EBMol <sub>930</sub>   | 99.35                        | 92.34                            | 96.88               | 86.61                 | 83.91                        | 45.68              |
| EBMol <sub>1370</sub>  | 99.65                        | 95.58                            | 98.32               | 82.78                 | 81.39                        | 43.37              |
| EBMol <sub>1810</sub>  | <b>99.76</b>                 | <b>97.09</b>                     | <b>98.93</b>        | 82.30                 | 81.42                        | 42.08              |

The results for unconditional generation on QM9 are summarized in Table 1. We benchmark EBMol for three different compute-settings, where the lowest setting of 930 network function evaluations (NFEs) matches the baseline models. The results show that EBMol<sub>930</sub> is competitive with the currently best bond-implicit model SLDM, while using approximately half the parameters.

When scaling compute to 1810 NFEs, EBMol exceeds all baselines in stability and validity. This includes the strong baseline GeoBFN that also achieves high quality samples through inference-time compute adaptation. Uniqueness and novelty are lower, at 82% and 42%, respectively. This is expected on QM9 because the dataset exhaustively enumerates its composition space [3], so higher-quality molecules are more likely to already appear in the training set and novel compositions tend to violate stability constraints [32, 40]. The quality–diversity tradeoff observed in Tables 1 and 2 is a direct consequence of the parallel tempering sampler concentrating probability mass on the lowest-energy modes. While this ensures high chemical validity, it naturally limits diversity.

Shifting towards the more challenging GEOM-Drugs benchmark, we report the revised metrics in Table 2. At a compute-matched setting, EBMol already surpasses the SLDM baseline, a performance gap that widens substantially as compute scales. While we observe a slight decrease in uniqueness as stability improves, this shift is consistent with the expected quality–diversity tradeoff. The model still produces 96.4% unique samples among 10k generations, suggesting a slow decay in diversity rather than a catastrophic mode collapse. Randomly selected samples are visualized in Appendix F.

The physical metrics presented in Table 3 clearly demonstrate the impact of the energy inductive bias. In the medium-compute configuration, EBMol<sub>1960</sub> already outperforms bond-implicit baselines and is the first such model to surpass the MMFF [55] force field. When compute is scaled, EBMol<sub>7240</sub> achieves a median energy decrease of 1.78 kcal/mol, an order of magnitude lower than previous bond-implicit models. Notably, EBMol remains competitive with the bond-explicit Megalodon (3.19 kcal/mol, 40.6M parameters), despite having 18 $\times$  fewer parameters and no explicit bond supervision. This trend extends to bond length and angle distributions, where EBMol<sub>7240</sub> matches or exceeds Megalodon’s performance.

## 4.3 Composable and conditional generation

**Shape-steered generation.** We compose the learned potential  $E_\theta$  with a differentiable potential  $U$  representing molecular shapes. This potential is defined via PCA eigenvalues of atom positions and targets either linear ( $U_{\text{lin}} = \lambda_2 + \lambda_3$ ), planar ( $U_{\text{disk}} = \lambda_3 + \lambda_1 - \lambda_2$ ) or spherical ( $U_{\text{sphere}} = \lambda_1 - \lambda_3$ ) shapes during generation. Here  $\lambda_1 \geq \lambda_2 \geq \lambda_3$  are the PCA eigenvalues of the covariance matrix of  $\mathbf{c}$ ,

Table 2: Revised stability, validity, uniqueness, novelty, and diversity metrics on GEOM-Drugs, computed on 10k generated samples per model. Compute budgets (NFEs) are indicated in each method’s subscript.

| Method                 | Atom $\uparrow$<br>Stability | Molecule $\uparrow$<br>Stability | Validity $\uparrow$ | Uniqueness $\uparrow$ | Novelty $\uparrow$ | Diversity $\uparrow$ |
|------------------------|------------------------------|----------------------------------|---------------------|-----------------------|--------------------|----------------------|
| Data                   | 99.60                        | 88.62                            | 97.91               | 97.89                 | –                  | 901.7                |
| Noise                  | 37.27                        | 00.00                            | 71.89               | 100.00                | 100.00             | 1873.8               |
| EDM <sub>1000</sub>    | 87.07                        | 1.03                             | 84.43               | <b>99.99</b>          | 99.94              | <b>1621.6</b>        |
| GEOLDM <sub>1000</sub> | 91.61                        | 6.08                             | 86.93               | 99.96                 | 99.88              | 1582.5               |
| SLDM <sub>1000</sub>   | 95.27                        | 20.59                            | 86.98               | 99.95                 | 99.93              | 1474.0               |
| EBMol <sub>1080</sub>  | 95.86                        | 24.68                            | 95.36               | 99.67                 | 99.69              | 1096.8               |
| EBMol <sub>1960</sub>  | 97.17                        | 37.40                            | 97.02               | 99.39                 | 99.50              | 927.0                |
| EBMol <sub>3720</sub>  | 98.15                        | 52.31                            | 98.61               | 98.47                 | 99.42              | 783.0                |
| EBMol <sub>7240</sub>  | <b>98.68</b>                 | <b>61.42</b>                     | <b>99.03</b>        | 96.42                 | 99.34              | 645.0                |

Table 3: Structural and energy metrics on GEOM-Drugs, computed on valid and connected molecules only. Bond lengths (Å), angles (degrees), energies (kcal/mol). The best bond-explicit models from GEOM-Drugs Revisited [52] are included for reference.

| Method                           | Bond Lengths $\downarrow$<br>( $\times 10^{-2}$ ) | Bond Angles $\downarrow$          | Torsions $\downarrow$             | Avg Energy $\downarrow$<br>Diff.  | Med Energy $\downarrow$<br>Diff.  |
|----------------------------------|---------------------------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| Data                             | $0.00 \pm 0.00$                                   | $0.00 \pm 0.00$                   | $0.00 \pm 0.01$                   | $0.00 \pm 0.01$                   | $0.00 \pm 0.00$                   |
| MMFF $\rightarrow$ GFN2-xTB      | $1.12 \pm 0.01$                                   | $1.22 \pm 0.00$                   | $4.89 \pm 0.10$                   | $11.40 \pm 0.20$                  | $9.84 \pm 0.06$                   |
| FlowMol2 <sub>100</sub> (4.3M)   | $1.30 \pm 0.04$                                   | $1.62 \pm 0.02$                   | $15.00 \pm 0.30$                  | $24.30 \pm 0.80$                  | $17.90 \pm 0.50$                  |
| Megalodon <sub>500</sub> (40.6M) | $0.66 \pm 0.02$                                   | $0.71 \pm 0.01$                   | $5.58 \pm 0.11$                   | $5.76 \pm 0.27$                   | $3.19 \pm 0.12$                   |
| EDM <sub>1000</sub> (2.2M)       | $5.26 \pm 0.03$                                   | $5.73 \pm 0.05$                   | $30.82 \pm 0.18$                  | $175.34 \pm 2.44$                 | $155.58 \pm 1.87$                 |
| GEOLDM <sub>1000</sub> (2.2M)    | $3.20 \pm 0.11$                                   | $3.90 \pm 0.06$                   | $24.51 \pm 0.19$                  | $99.25 \pm 4.49$                  | $80.13 \pm 2.13$                  |
| SLDM <sub>1000</sub> (4.2M)      | $1.66 \pm 0.03$                                   | $1.34 \pm 0.02$                   | $10.07 \pm 0.08$                  | $23.77 \pm 0.87$                  | $17.90 \pm 0.30$                  |
| EBMol <sub>1080</sub> (2.2M)     | $1.77 \pm 0.02$                                   | $1.67 \pm 0.03$                   | $8.46 \pm 0.13$                   | $40.77 \pm 9.16$                  | $6.67 \pm 0.30$                   |
| EBMol <sub>1960</sub> (2.2M)     | $1.22 \pm 0.04$                                   | $1.19 \pm 0.04$                   | $6.53 \pm 0.19$                   | $26.13 \pm 7.87$                  | $4.12 \pm 0.07$                   |
| EBMol <sub>3720</sub> (2.2M)     | $0.83 \pm 0.01$                                   | $0.81 \pm 0.01$                   | $4.56 \pm 0.07$                   | $10.36 \pm 1.55$                  | $2.35 \pm 0.05$                   |
| EBMol <sub>7240</sub> (2.2M)     | <b><math>0.65 \pm 0.01</math></b>                 | <b><math>0.62 \pm 0.01</math></b> | <b><math>3.65 \pm 0.06</math></b> | <b><math>5.27 \pm 0.39</math></b> | <b><math>1.78 \pm 0.04</math></b> |

measuring the stretch along each principal axis. For example, minimizing  $U_{\text{lin}} = \lambda_2 + \lambda_3$  suppresses the two minor axes, thus collapsing the molecule onto a single axis. Steered molecules successfully shift toward the target shape (Figure 2) while maintaining reasonable stability and validity (Table 4), confirming that the learned energy landscape contains valid basins even for shapes underrepresented in the training data.

**Case study on linker design.** The additive energy composition naturally enables conditioning on arbitrary inputs, without any retraining. To test this, we conduct a linker design case study where

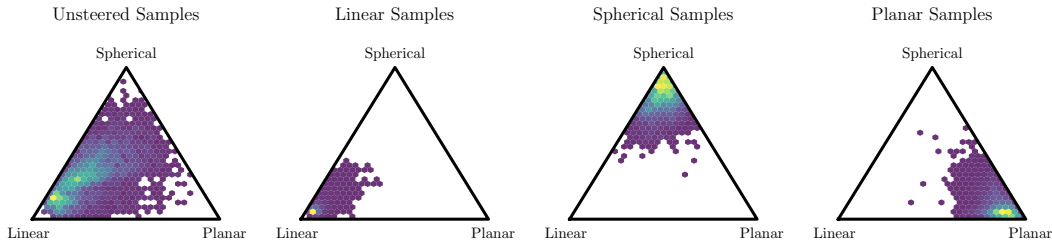


Figure 2: Shape distribution of 5000 generated molecules. Left: unsteered generation. Right: generation with composable shape potentials targeting linear, planar, and spherical geometries. Steering successfully redirects sampling toward underrepresented shapes.



Table 4: Zero-shot conditional generation results. **Left:** Linker design across six problems of varying complexity, each evaluated over 100 attempts. Success requires a single connected component passing RDKit sanitization. **Right:** Shape-steered generation across three target shape classes alongside unsteered generation, each over 5000 samples. No task-specific training is required.

| <i>Linker design</i> |                   |                                    |                                    | <i>Shape-steered generation</i> |                                         |                                         |                                       |
|----------------------|-------------------|------------------------------------|------------------------------------|---------------------------------|-----------------------------------------|-----------------------------------------|---------------------------------------|
| <b>Problem</b>       | <b>Free Atoms</b> | <b>Succ. <math>\uparrow</math></b> | <b>Uniq. <math>\uparrow</math></b> | <b>Target</b>                   | <b>Mol. Stab. <math>\uparrow</math></b> | <b>Bond Stab. <math>\uparrow</math></b> | <b>Validity <math>\uparrow</math></b> |
| P1 (easy)            | 5                 | 95.2                               | 16.4                               | Linear                          | 20.76                                   | 95.57                                   | 88.28                                 |
| P2 (medium)          | 7                 | 77.9                               | 48.1                               | Planar                          | 8.96                                    | 92.90                                   | 88.80                                 |
| P3 (medium)          | 12                | 88.5                               | 84.5                               | Spherical                       | 31.66                                   | 96.59                                   | 96.70                                 |
| P4 (hard, 3 frags)   | 10                | 55.8                               | 51.5                               |                                 |                                         |                                         |                                       |
| P5 (hard, 2 locs)    | 5 + 5             | 75.0                               | 83.3                               | Unsteered                       | 37.52                                   | 97.09                                   | 96.68                                 |
| P6 (hard, 2 locs)    | 8 + 6             | 51.0                               | 93.3                               |                                 |                                         |                                         |                                       |
| Average              |                   | 73.9                               | 62.85                              |                                 |                                         |                                         |                                       |

we fix a subset of atoms and sample the remaining ones. We evaluate this zero-shot linker design by removing linking fragments between stable parts of molecules like ring structures and replacing the removed atoms with Gaussian noise matched to the removed atoms spatial statistics. We subsequently run a modified parallel tempering sampler for 2000 iterations. Following the evaluation protocol of [56], EBMol achieves an average success rate of 73.9% across six conditioning tasks of varying complexity (Table 4), all without task-specific training.

#### 4.4 Physical meaningfulness of the learned energy

The energy inductive bias not only improves generation quality but produces a physically meaningful scoring function. We evaluate the correlation between EBMol energy and the energy decrease upon GFN2-xTB [54] relaxation for molecules generated by the independently trained EDM (Figure 6 in Appendix C). The correlation ( $r = 0.65$ ,  $p < 0.0001$ ) emerges despite EBMol never receiving any energy-related supervision: the model is trained solely on atomic positions and atom types, yet learns to assign high energy to physically strained geometries. As a result,  $E_\theta$  can serve as a lightweight first-pass filter for screening generated molecules by physical quality, providing order-of-magnitude speedups over GFN2-xTB optimization when screening large molecular batches.

## 5 Conclusion and limitations

**Contributions.** EBMol demonstrates the potential of energy-based models to serve as a versatile and practical framework for 3D molecular generation, achieving state-of-the-art results on QM9 and GEOM-Drugs while providing critical capabilities that standard distribution-matching frameworks lack: energy-based ranking, steering via potential composition, and conditional generation from a single model. The core components are Restoring Field Matching, which shapes a scalar energy landscape without simulation, and Mirror-Langevin sampling, unifying continuous and discrete updates within a principled framework. Perhaps most notably, EBMol generates structures surprisingly close to physical local minima: its median relaxation energy of 1.78 kcal/mol is an order of magnitude below previous bond-implicit methods. This performance is competitive with bond-explicit models at  $18\times$  the parameter count, despite EBMol requiring no bond or energy supervision.

**Limitations.** While we establish the design space for learning an energy landscape combined with Mirror-Langevin sampling, we do not systematically explore advanced sampler designs beyond the three algorithms in our sampler ablation (Table 7). Currently, achieving the best results requires a significant number of NFEs, and the employed sampling algorithms are sensitive to hyperparameter choices. On the theoretical side, our analysis establishes well-posedness of the training problem and characterizes the solution at the regularization extremes, but the claim that data points correspond to local minima in the learned landscape is motivated by the target construction and verified only empirically.

**Outlook.** A natural extension to EBMol is integrating energy calibration by replacing the zero-anchor objective with a loss that aligns  $E_\theta(\mathbf{x}_0)$  to a physical quantity such as DFT or GFN2-xTB energy values. A calibrated energy would also enable coupled generative-simulation workflows in which the same  $E_\theta$  both generates candidate molecules and serves as an approximate molecular dynamics potential. On the sampling side, we expect significant improvements from more sophisticated samplers such as accelerated Langevin methods [57, 58] for faster sampling or from repulsive interaction potentials [17] for improved sample diversity.

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## A Proofs

In this section we provide rigorous formulations and proofs of the theoretical results in section 3.5. Let us first fix some notation. We denote the marginal densities of  $\mathbf{x}_t$  by  $p_t = \text{law}(\mathbf{x}_t)$  and  $p_x(\mathbf{x}) = \int p_t(\mathbf{x}) p(t) dt$ . We denote the domain of  $\mathbf{x}$  as  $\Omega = \mathbb{R}^{N \times 3} \times \Delta^{K-1}$ . We define the Sobolev space  $W^{1,2}(\Omega) = \{f \in L^2(\Omega) \mid \nabla f \in L^2(\Omega)\}$  where the gradient is understood in the usual weak sense. Moreover, as the underlying solution space for our training problem we define the weighted Sobolev space

$$\mathcal{W}^{1,2}(\Omega) = \left\{ f : \Omega \rightarrow \mathbb{R} \mid \nabla f \text{ exists weakly and } \int_{\Omega} (|f(\mathbf{x})|^2 + |\nabla f(\mathbf{x})|^2) p_x(\mathbf{x}) d\mathbf{x} < \infty \right\}. \quad (10)$$

We impose the following assumptions on the occurring distributions, respectively their densities.

### Assumption 1.

1. The domain  $\text{dom}(p_x)$  is connected,  $p_x$  is continuous, bounded above and strictly positive on  $\Omega$ . Moreover,  $p_x$  satisfies a Poincaré inequality, that is, there exists  $C > 0$  such that

$$\|f - \bar{f}\|_{L^2(p_x)} \leq C \|\nabla f\|_{L^2(p_x)}, \quad f \in \mathcal{W}^{1,2}(\Omega) \quad (\text{PI})$$

where  $\bar{f} = \int f(\mathbf{x}) p_x(\mathbf{x}) d\mathbf{x}$ .

2. It holds  $p_0 \ll p_x$  with  $\frac{p_0}{p_x}$  locally bounded. Moreover, there exists  $A \subset \mathbb{R}^d$  such that for  $\mathbf{x} \in A$ ,  $p_0(\mathbf{x}) = 0$  implies  $p_x(\mathbf{x}) = 0$  and  $\int_A p_x(\mathbf{x}) d\mathbf{x} > 0$ .

**Remark 1.** Let us briefly comment on assumption 1.

1. In the first assumption the most interesting part is the Poincaré inequality. This assumption appears frequently in the context of diffusion processes and Langevin sampling [59]. A Poincaré inequality is, in particular, satisfied for a density  $p$ , if

$$\langle \nabla \log p(\mathbf{x}), \mathbf{x} \rangle \leq -a \|\mathbf{x}\| + b$$

for some  $a > 0$ ,  $b \in \mathbb{R}$  [60]. That is, the Poincaré inequality imposes in some sense sufficiently fast decaying tails on the density.

2. Regarding the second assumption,  $p_0 \ll p_x$  is to be expected, as  $p_x$  is effectively a perturbed version of  $p_0$ . The converse absolute continuity on the set  $A$  is, for instance, satisfied if there exists a set  $A$  of positive measure  $p_0 > 0$  on  $A$ .

Before considering the main results, we prove the following equivalence of norms in  $\mathcal{W}^{1,2}(\Omega)$ .

**Lemma 1.** *Under assumption 1 for every  $u \in \mathcal{W}^{1,2}(\Omega)$  we have*

$$\|u\|_{L^2(p_x)} \leq \|u\|_{L^2(p_0)} + \|\nabla u\|_{L^2(p_x)}. \quad (11)$$

**Remark 2.** *The important aspect above is that the first  $L^2$  norm on the right-hand side is with respect to  $p_0$ .*

*Proof.* Assume to the contrary, we find a sequence  $(u_n)_n \subset \mathcal{W}^{1,2}(\Omega)$  with

$$\|u_n\|_{L^2(p_x)} \geq n(\|u_n\|_{L^2(p_0)} + \|\nabla u_n\|_{L^2(p_x)}). \quad (12)$$

Define

$$v_n = \frac{u_n}{\|u_n\|_{L^2(p_x)}} \quad (13)$$

so that  $\|v_n\|_{L^2(p_x)} = 1$  and by (12)

$$\frac{1}{n} \geq (\|v_n\|_{L^2(p_0)} + \|\nabla v_n\|_{L^2(p_x)}). \quad (14)$$

In particular, we can extract a subsequence  $(v_{n_k})_k$  such that

$$\begin{aligned} v_{n_k} &\rightharpoonup v \\ \nabla v_{n_k} &\rightharpoonup g \end{aligned} \quad (15)$$

both in  $L^2(p_x)$  and by the assumed positivity and continuity of  $p_x$  we find in particular, that  $v_{n_k} \rightharpoonup v$  in  $W_{\text{loc}}^{1,2}(\Omega)$ , that is, with respect to the regular Lebesgue measure, and  $g = \nabla v$ . Moreover,  $v \in \mathcal{W}^{1,2}(\Omega)$ . Now, (14) implies that  $\nabla v_{n_k} \rightarrow 0$  and since weak limits are unique we have  $\nabla v = g = 0$ . The Poincaré inequality, cf. assumption 1, then implies

$$\|v - \bar{v}\|_{L^2(p_x)} \leq C\|\nabla v\|_{L^2(p_x)} = 0,$$

that is,  $v(\mathbf{x})$  is equal to a constant, say  $c \in \mathbb{R}$ ,  $p_x$ -a.e. Similarly, (14) yields  $v_{n_k} \rightarrow 0$  in  $L^2(p_0)$  so that  $v(\mathbf{x}) = 0$  for  $p_0$ -a.e.  $\mathbf{x}$ . Using the set  $A$  from assumption 1 we find

$$c \int_A p_x(\mathbf{x}) d\mathbf{x} = \int_A v(\mathbf{x}) \frac{p_x(\mathbf{x})}{p_0(\mathbf{x})} p_0(\mathbf{x}) d\mathbf{x} = 0 \quad (16)$$

where the last equality follows from  $v(\mathbf{x}) = 0$ ,  $p_0$ -a.e. Since  $\int_A p_x(\mathbf{x}) d\mathbf{x} > 0$  it follows  $c = 0$ , that is,  $v = 0$ ,  $p_x$ -a.e. Thus, we have that  $v_{n_k} \rightarrow 0$  in  $L^2(p_x)$ . Since constants are elements of  $L^2(p_x)$  this also implies that

$$\bar{v}_{n_k} = \int v_{n_k}(\mathbf{x}) p_x(\mathbf{x}) d\mathbf{x} \rightarrow 0. \quad (17)$$

Then, the Poincaré inequality implies

$$\begin{aligned} \|v_{n_k}\|_{L^2(p_x)} &\leq \|v_{n_k} - \bar{v}_{n_k}\|_{L^2(p_x)} + \|\bar{v}_{n_k}\|_{L^2(p_x)} \\ &\leq \|v_{n_k} - \bar{v}_{n_k}\|_{L^2(p_x)} + \bar{v}_{n_k} \\ &\leq C\|\nabla v_{n_k}\|_{L^2(p_x)} + \bar{v}_{n_k} \rightarrow 0 \end{aligned} \quad (18)$$

We, therefore, find

$$1 = \|v_{n_k}\|_{L^2(p_x)} \rightarrow 0 \quad (19)$$

which is a contradiction.  $\square$

We can now prove existence and uniqueness of solutions of the training problem. Let us first rigorously reformulate the result.

**Proposition 3** (Existence of solutions). *Under assumption 1 the problem*

$$\min_{E \in \mathcal{W}^{1,2}(\Omega)} \mathcal{L}(E)$$

*admits a unique solution.*

*Proof.* Denoting  $\mathbf{u} = (\mathbf{u}^c, \mathbf{u}^p)$  we can rewrite the objective as

$$\begin{aligned}\mathcal{L}(E) &= \iiint \left\{ \left\| -\nabla E_\theta(\mathbf{x}_t) - \mathbf{u} \right\|^2 + \lambda_{\text{reg}} E(\mathbf{x}_0)^2 \right\} p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt \\ &= \iiint \left\| -\nabla E_\theta(\mathbf{x}_t) - \mathbf{u} \right\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt + \lambda_{\text{reg}} \int E(\mathbf{x})^2 p_0(\mathbf{x}) d\mathbf{x}\end{aligned}\quad (20)$$

Now let  $(E_n)_n \subset \mathcal{W}^{1,2}(\Omega)$  be a minimizing sequence, that is,

$$\lim_{n \rightarrow \infty} \mathcal{L}(E_n) = \inf_{E \in \mathcal{W}^{1,2}(\Omega)} \mathcal{L}(E).$$

By definition of  $\mathcal{L}$  it follows that  $\|E_n\|_{L^2(p_0)}$  and  $\|\nabla E_n\|_{L^2(p_x)}$  are bounded. The Poincaré inequality, cf. lemma 1 then yields that also  $\|E_n\|_{L^2(p_x)}$  is bounded so that we can extract a subsequence  $(E_{n_k})_k$  and functions  $E^* \in L^2(p_x)$ ,  $g \in L^2(p_x)$  such that  $E_{n_k} \rightharpoonup E^*$ ,  $\nabla E_{n_k} \rightharpoonup g$  both in  $L^2(p_x)$ . Moreover, by the assumed properties of  $p_x$  we also find that the weak convergence is in  $W^{1,2}(\Omega)$  and that  $\nabla E^* = g$  and  $E^* \in \mathcal{W}^{1,2}(\Omega)$ . By potentially extracting another subsequence,  $E_{n_k}$  converges also weakly in  $L^2(p_0)$  and the limit is again  $E^*$  due to the assumed boundedness of  $p_0/p_x$  as the latter implies that weak convergence in  $L^2(p_x)$  includes also weak convergence in  $L^2(p_0)$ . Note that

$$G \mapsto \iiint \left\| -G(\mathbf{x}_t) - \mathbf{u} \right\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt. \quad (21)$$

is convex and continuous in  $L^2(p_x)$  and therefore also lower semi-continuous with respect to weak convergence in  $L^2(p_x)$ . Similarly,

$$E \mapsto \iiint \lambda_{\text{reg}} E(\mathbf{x}_0)^2 p(\mathbf{x}_0) d\mathbf{x}_0 \quad (22)$$

is weakly lower semi-continuous in  $L^2(p_0)$ . Thus,

$$\mathcal{L}(E^*) \leq \liminf_{n \rightarrow \infty} \mathcal{L}(E_n) = \inf_E \mathcal{L}(E) \quad (23)$$

showing existence of a solution. Uniqueness is a consequence of strict convexity of the objective.  $\square$

Having established well-posedness we can consider the concise formulation and proof of proposition 2.

**Proposition 4** (Convergence for small and large regularization parameters). *Under assumption 1, let  $E^*(x, \lambda)$  be the minimizer of  $\mathcal{L}$  for  $\lambda_{\text{reg}} = \lambda$ . It holds true that*

- (i) *There exists a weak accumulation point  $E_\infty^* \in \mathcal{W}^{1,2}(\Omega)$  of  $E^*(\cdot, \lambda)$  as  $\lambda \rightarrow \infty$  and every such accumulation point satisfies  $E_\infty^*(x) = 0$  for  $p_0$ -a.e.  $x$ .*
- (ii) *There exists  $E_0^* \in \mathcal{W}^{1,2}(\Omega)$  such that  $\nabla E_0^*$  is a weak accumulation point of  $\nabla E^*(\cdot, \lambda)$  as  $\lambda \rightarrow 0$  and every such  $E_0^*$  minimizes*

$$\mathcal{W}^{1,2}(\Omega) \ni G \mapsto \iiint \left\| -\nabla G(\mathbf{x}_t) - \mathbf{u} \right\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt, \quad (24)$$

*that is,  $\nabla E_0^*$  is precisely the projection of  $\mathbf{u}$  onto the space of all gradient fields. Moreover, if there exists  $F \in \mathcal{W}^{1,2}(\Omega)$  with*

$$\nabla F(x) = -\mathbb{E}[\mathbf{u} | \mathbf{x}_t = \mathbf{x}], \quad p_x - \text{a.e. } \mathbf{x}. \quad (25)$$

*then  $\nabla E^*(\mathbf{x}) = -\mathbb{E}[\mathbf{u} | \mathbf{x}_t = \mathbf{x}]$ ,  $p_x$ -a.e.*

- (iii) *If, in addition it holds true that  $v(\mathbf{x}_0, \mathbf{x}_1, t) = v(\mathbf{x}_t)$ , that is,  $v$  depends solely on  $\mathbf{x}_t$ , then there exists a weak accumulation point  $E_0^* \in \mathcal{W}^{1,2}(\Omega)$  of  $E^*(\cdot, \lambda)$  as  $\lambda \rightarrow 0$  and every such accumulation point satisfies  $\nabla E_0^* = -\mathbf{u}$ ,  $p_x$ -a.e.*

*Proof.* We begin with the limit for  $\lambda \rightarrow \infty$ . Assume without loss of generality  $\lambda \geq 1$ . By optimality of  $E^*(\cdot, \lambda)$

$$\begin{aligned}
& \int E^*(\mathbf{x}_0, \lambda)^2 p_0(\mathbf{x}_0) d\mathbf{x}_0 \\
& \leq \iiint \|\nabla E^*(\mathbf{x}_t, \lambda) - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt + \lambda \int E^*(\mathbf{x}_0, \lambda)^2 p_0(\mathbf{x}_0) d\mathbf{x}_0 \\
& \leq \iiint \|0 - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt + \lambda \int 0 d\mathbf{x}_0 \\
& \leq \iiint \|\mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt < \infty
\end{aligned} \tag{26}$$

which implies that  $\|E^*(\cdot, \lambda)\|_{L^2(p_0)}$  and  $\|\nabla E^*(\cdot, \lambda)\|_{L^2(p_x)}$  are bounded for all  $\lambda \geq 1$ . As above, we can therefore extract a weakly convergent subsequence  $E^*(\cdot, \lambda_n) \rightharpoonup E_\infty^*$  in  $\mathcal{W}^{1,2}(\Omega)$  as well as in  $L^2(p_0)$ . By weak lower semi-continuity of  $E \mapsto \int E(\mathbf{x})^2 p_0(\mathbf{x}) d\mathbf{x}$  and (26) it follows

$$\begin{aligned}
\int E_\infty^*(\mathbf{x}_0)^2 p_0(\mathbf{x}_0) d\mathbf{x}_0 & \leq \liminf_{n \rightarrow \infty} \int E^*(\mathbf{x}_0, \lambda_n)^2 p_0(\mathbf{x}_0) d\mathbf{x}_0 \\
& \leq \frac{1}{\lambda_n} \iiint \|\mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt \rightarrow 0
\end{aligned} \tag{27}$$

as  $n \rightarrow \infty$  implying  $E_\infty^* = 0$ ,  $p(\mathbf{x}_0)$ -a.e.

Regarding the convergence as  $\lambda \rightarrow 0$ , as above we find that

$$\begin{aligned}
& \iiint \|\nabla E^*(\mathbf{x}_t, \lambda) - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt \\
& \leq \iiint \|\mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt < \infty.
\end{aligned} \tag{28}$$

Thus, we can only deduce boundedness of  $\nabla E^*(\cdot, \lambda)$  in  $L^2(p_x)$  as  $\lambda \rightarrow 0$ . By the Poincaré inequality (cf. assumption 1), however, we also have that  $F^*(\cdot, \lambda) = E^*(\cdot, \lambda) - \bar{E}^*(\lambda)$  is bounded in  $L^2(p_x)$ . Therefore, we may extract a subsequence such that  $F^*(\cdot, \lambda_n) \rightharpoonup F_0^*$  as  $n \rightarrow \infty$ . In particular, it follows that  $\nabla E^*(\cdot, \lambda_n) = \nabla F^*(\cdot, \lambda_n) \rightharpoonup \nabla F_0^*$ . Once again, by weak lower semicontinuity and optimality we find for any  $G \in \mathcal{W}^{1,2}(\Omega)$

$$\begin{aligned}
& \iiint \|\nabla F_0^*(\mathbf{x}_t) - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt \\
& \leq \liminf_{n \rightarrow \infty} \iiint \|\nabla E^*(\mathbf{x}_t, \lambda_n) - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt \\
& \leq \liminf_{n \rightarrow \infty} \iiint \|\nabla E^*(\mathbf{x}_t, \lambda_n) - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt + \lambda_n \int E^*(\mathbf{x}_0, \lambda)^2 p(\mathbf{x}_0) d\mathbf{x}_0 \\
& \leq \liminf_{n \rightarrow \infty} \iiint \|\nabla G(\mathbf{x}_t) - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt + \lambda_n \int G(\mathbf{x}_0)^2 p(\mathbf{x}_0) d\mathbf{x}_0 \\
& \leq \iiint \|\nabla G(\mathbf{x}_t) - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt
\end{aligned} \tag{29}$$

implying the desired result. The representation,  $\nabla E^*(\mathbf{x}) = -\mathbb{E}[\mathbf{u} | \mathbf{x}_t = \mathbf{x}]$  follows directly by the fact that

$$w \mapsto \iiint \|w(\mathbf{x}_t) - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt \tag{30}$$

is minimized for  $w(\mathbf{x}) = \mathbb{E}[\mathbf{u} | \mathbf{x}_t = \mathbf{x}]$ .

In the last case, we note that

$$\begin{aligned}
& \iiint \|w(\mathbf{x}_t) - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt \\
& = \int \left( \iint \|w(\mathbf{x}_t) - \mathbf{u}\|^2 p(\mathbf{x}_0, t | \mathbf{x}_t) d\mathbf{x}_0 dt \right) p(\mathbf{x}_t) d\mathbf{x}_t
\end{aligned} \tag{31}$$



which is zero for  $w(\mathbf{x}_t) = \mathbf{u}(\mathbf{x}_t)$ . Therefore, under the assumption that there exists  $F \in \mathcal{W}^{1,2}(\Omega)$  with  $\nabla F(\mathbf{x}_t) = \mathbf{u}(\mathbf{x}_t)$ , we find by optimality of  $E^*(\cdot, \lambda)$

$$\begin{aligned} & \iiint \|\nabla E^*(\mathbf{x}_t, \lambda) - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt + \lambda_n \int E^*(\mathbf{x}_0, \lambda)^2 p_0(\mathbf{x}_0) d\mathbf{x}_0 \\ & \leq \iiint \|\nabla F(\mathbf{x}_t) - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt + \lambda_n \int F(\mathbf{x}_0)^2 p_0(\mathbf{x}_0) d\mathbf{x}_0 \\ & = \lambda_n \int F(\mathbf{x}_0)^2 p_0(\mathbf{x}_0) d\mathbf{x}_0 \end{aligned} \quad (32)$$

so that in this case obtain again boundedness of  $\nabla E^*(\cdot, \lambda)$  and  $E^*(\cdot, \lambda)$  both in  $L^2(p_x)$ . Extracting weakly convergent subsequences again with limit  $E_0^*$  we find by weak lower semi-continuity

$$\begin{aligned} & \iiint \|\nabla E_0^*(\mathbf{x}_t) - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt \\ & \leq \liminf_{n \rightarrow \infty} \iiint \|\nabla E^*(\mathbf{x}_t, \lambda_n) - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt \\ & \leq \liminf_{n \rightarrow \infty} \iiint \|\nabla E^*(\mathbf{x}_t, \lambda_n) - \mathbf{u}\|^2 p(\mathbf{x}_0, \mathbf{x}_t, t) d\mathbf{x}_t d\mathbf{x}_0 dt + \lambda_n \int E^*(\mathbf{x}_0, \lambda)^2 p_0(\mathbf{x}_0) d\mathbf{x}_0 \\ & \leq \liminf_{n \rightarrow \infty} \lambda_n \int F(\mathbf{x}_0, \lambda)^2 p_0(\mathbf{x}_0) d\mathbf{x}_0 = 0 \end{aligned} \quad (33)$$

concluding the proof.  $\square$

## B Empirical Verification of Local Minima

The RFM construction aims to place data points at local minima of the learned energy, as the smoothing function ensures pointwise stationarity at  $t = 0$ , and the symmetric restoring targets induce non-negative curvature around data. The theoretical analysis in Section 3.5 establishes existence and uniqueness of the loss minimizer and characterizes its behavior under varying regularization strength, but does not formally prove the local-minima property for the marginalized predictor. Here we verify empirically that the trained network  $E_\theta$  realizes this intended behavior through three complementary experiments.

**Deterministic gradient relaxation from data.** We initialize 1000 molecules taken from the training and test sets and run deterministic gradient descent ( $\tau = 0$ ) for  $N = 500$  steps with step size  $\eta = 0.01$  on the learned energy landscape. We evaluate atom displacement via mean RMSD, the energy change during relaxation, and the change in molecule stability. We conduct this experiment for two models: EBMol trained with the full RFM objective, and the OTFM ablation from Table 6, which is trained without the extrapolation construction.

Table 5: Deterministic gradient relaxation initialized at data. Energy values, RMSD (in  $\text{\AA}$ ), and stability after  $N$  steps of gradient descent on  $E_\theta$ .

| Method      | Split | Median $\Delta E \downarrow$ | Mean RMSD $\downarrow$ | $\Delta$ Mol. Stability $\downarrow$ |
|-------------|-------|------------------------------|------------------------|--------------------------------------|
| EBMol (RFM) | Train | -0.59                        | 0.121                  | 2.4                                  |
| EBMol (RFM) | Test  | -0.67                        | 0.124                  | 2.1                                  |
| OTFM (6)    | Train | -154.98                      | 143.365                | 63.1                                 |

Results are reported in Table 5. Three observations confirm the role of the RFM construction. First, on training data, EBMol configurations remain effectively stationary (RMSD 0.121, stability loss 2.4 percentage points), verifying that the RFM loss produces energy minima at training samples as intended. Second, the same holds on held-out test data (RMSD 0.124, stability loss 2.1 percentage points), demonstrating that the local-minima property generalizes to molecules unseen during training. Third, under the OTFM loss, configurations diverge dramatically (RMSD 143.4) into spurious minima that do not correspond to valid molecular structures, with molecule stability dropping by 63 percentage points. This confirms that the extrapolation construction of RFM is responsible for placing data at local minima.

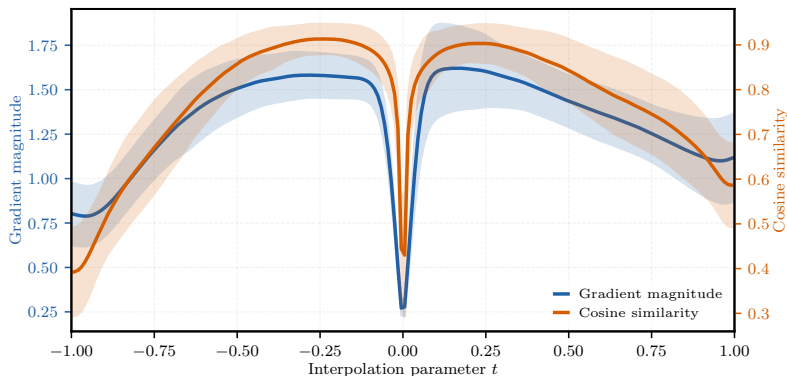


Figure 3: Gradient (wrt. position) magnitude (blue) and cosine similarity between the learned restoring field and the prescribed target direction (orange) along interpolation paths from prior ( $t = \pm 1$ ) through data ( $t = 0$ ). Shaded regions indicate one standard deviation over 1000 molecules. The gradient vanishes near data, confirming stationarity, while the high cosine similarity away from data confirms that the learned field consistently points toward training molecules.

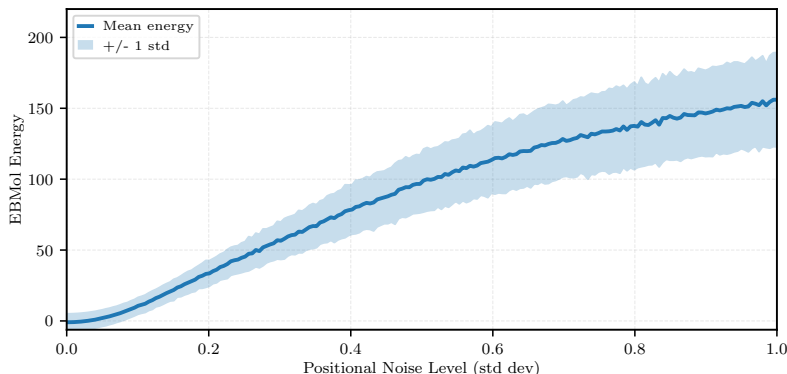


Figure 4: Learned energy  $E_\theta$  as a function of Gaussian noise magnitude  $\sigma$  applied to training molecule coordinates. Shaded regions indicate one standard deviation over 1000 molecules. The monotonic increase confirms that data points occupy local minima of the learned landscape.

**Energy and gradient profiles along interpolation paths.** We evaluate the gradient magnitude with respect to the position  $\|\nabla_c E_\theta(\mathbf{x}_t)\|$  and the cosine similarity between  $\nabla_c E_\theta(\mathbf{x}_t)$  and the unsmoothed restoring direction  $\mathbf{u}^c = \text{sign}(t) \cdot (\mathbf{c}_0 - \mathbf{c}_1)$ , along linear interpolation paths between training molecules ( $t = 0$ ) and corresponding optimal-transport-matched prior samples ( $t = \pm 1$ ).

Figure 3 shows the results averaged over 1000 molecules. The gradient magnitude drops sharply near  $t = 0$ , consistent with data sitting at stationary points of the learned energy. Away from data, the gradient grows monotonically toward the prior, indicating increasingly steep energy walls around each basin. The cosine similarity remains high ( $> 0.6$ ) throughout most of the interval, confirming that the learned gradient field points back toward data across the interpolation range, and drops near  $t = 0$  where the vanishing gradient magnitude makes the direction ill-defined. The symmetric profile across  $t < 0$  and  $t > 0$  reflects the symmetric construction of the RFM target.

**Response to Gaussian perturbation.** We perturb training molecules by adding isotropic Gaussian noise  $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$  to atomic coordinates for increasing noise levels  $\sigma$  and evaluate  $E_\theta$  at the perturbed configurations. Figure 4 shows the learned energy as a function of  $\sigma$ , averaged over 1000 molecules. The energy increases smoothly and monotonically with perturbation magnitude, starting near zero at clean data. This confirms that data points sit at the center of well-shaped energy basins, with the learned landscape assigning progressively higher energy to configurations that deviate from valid molecular geometries.

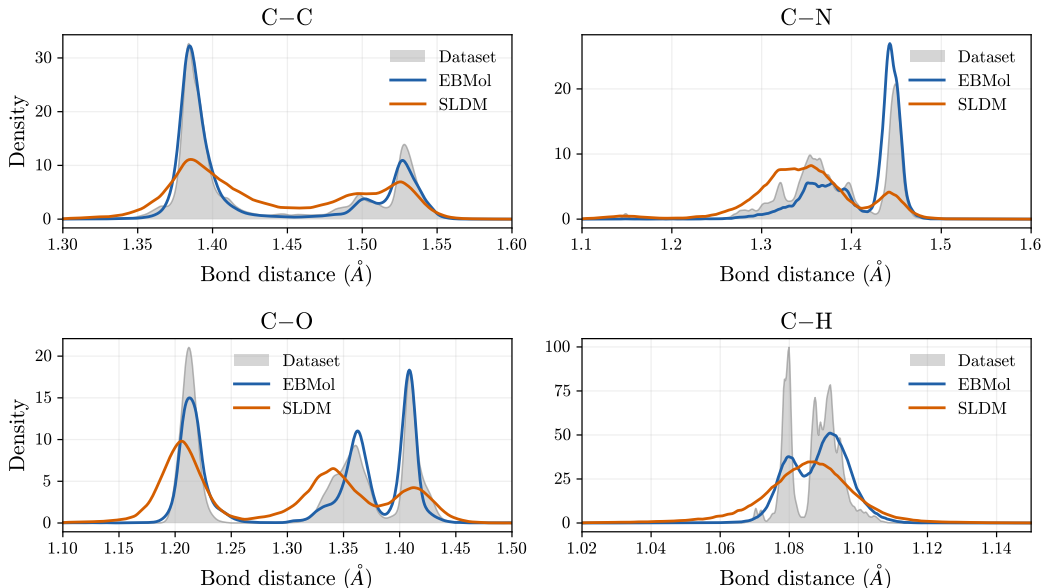


Figure 5: Bond distance distributions for C–C, C–N, C–O, and C–H pairs on GEOM-Drugs, comparing EBMol<sub>7240</sub> (blue) and SLDM (orange) against the dataset (gray). Distinct peaks correspond to different bond orders. EBMol recovers the multimodal structure of the reference distributions, with sharper peak separation and more realistic proportions than SLDM.

## C Structural and energetic analysis

**Bond distance distributions.** Figure 5 provides a per-element decomposition of the aggregate bond length errors reported in Table 3, comparing bond distance distributions for molecules generated by SLDM and EBMol<sub>7240</sub> against the training data on GEOM-Drugs. EBMol recovers the C–C distribution with high fidelity, accurately capturing peak locations, relative heights, and the separation between single, aromatic, and double bond modes. For C–N and C–O, EBMol correctly resolves relative peak proportions, which SLDM fails to achieve. EBMol also successfully recovers the bimodality of C–H bonds.

**Cross-model energy correlation.** Figure 6 plots EBMol’s learned energy against the energy decrease upon GFN2-xTB relaxation for 10,000 molecules generated by EDM. The positive correlation and its interpretation are discussed in Section 4.4.

## D Training objective and sampler ablation

**Training objective ablation.** A learned energy landscape requires a sampler to generate samples. We therefore evaluate each training objective under three different samplers (Forward Euler (FWDE), Annealed Langevin Dynamics (ALD), and Parallel Tempering (PT)) to disentangle the contributions of training and sampling. All ablations use a reduced-size model with embedding dimension halved to 128 to keep the search tractable. To ensure a fair comparison, we tune sampler hyperparameters separately for each training objective. For FWDE, we sweep the step size over  $[0.05, 0.001]$  and select the configuration with the highest atom stability. For the FM baselines this controls how finely the  $[0, 1]$  integration interval is discretized, giving them the best possible performance under their natural sampling procedure. For ALD, we lock in the selected step size and additionally sweep over the noise scale  $\sigma$ , using the same value for coordinates and atom types for simplicity. For PT, we reuse the ALD configuration. We acknowledge this sequential scheme is suboptimal but it keeps the search feasible across all method-sampler combinations. All experiments use a fixed budget of 1080 NFE per sample on average and  $n = 500$  samples per experiment. In addition to atom stability and validity, we report two metrics specific to this ablation: W2, the Wasserstein-2 distance between

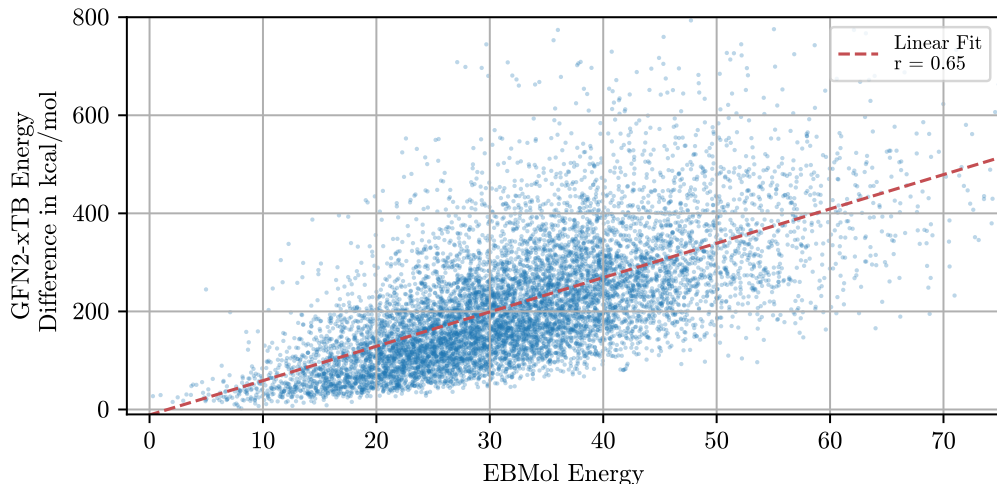


Figure 6: The learned energy correlates with physical quality across models. EBMol energy vs. energy decrease upon GFN2-xTB relaxation for 10k molecules generated by EDM ( $r = 0.65$ ,  $p < 0.0001$ ). Higher EBMol energy predicts larger relaxation energy, indicating physically strained geometries.

pairwise atom distance distributions of generated and reference molecules restricted to the 1.0–2.0 Å range most relevant for bond lengths, and the number of diverged samples that exceeded plausible energy bounds during sampling and were discarded. Results are reported in Table 6.

We ablate against two baselines that share EBMol’s energy architecture but are trained with the flow matching loss: Standard FM (without optimal transport or atom permutation alignment) and Equivariant OT-FM (with SE(3)-equivariant optimal transport coupling and atom permutation). The gap between the two is substantial, confirming that equivariant OT alignment is critical for learning a useful energy landscape in our framework. However, Equivariant OT-FM produces a significant number of diverged samples across all samplers, consistent with the relaxation experiments in Section B: without the RFM construction, data points are not local minima, and gradient-based sampling can escape into unphysical regions. This problem is most severe under PT, where high-temperature chains actively encourage exploration of the landscape. We note that the 1074 diverged samples for only 500 generated samples is possible because our parallel tempering implementation replaces diverged samples with fresh prior samples periodically. This also precludes inference-time compute scaling, since increasing the sampling budget leads to more divergences rather than better samples.

The RFM block tells a different story. No RFM variant produces a single diverged sample, confirming that the restoring-field construction yields a landscape that is globally stable under sampling. Within the RFM variants, several trends emerge. Comparing RFM with and without the smoothing function  $b(t)$  shows that smoothing improves the W2 bond-distance metric, suggesting that the smoother basins are easier for the sampler to navigate, though PT partially compensates for the absence of smoothing. Moving from a static isotropic prior to the data-driven anisotropic prior described in the Implementation section yields a further improvement, most visible in the Valid & Connected rate: the full RFM model achieves 37.7% under PT, compared to 22.6% for the static-prior variant. This indicates that the data-driven prior helps the sampler to find not just locally valid atomic environments (atom stability) but globally coherent molecular structures (validity and connectivity).

We ablate sampler choices for our full RFM model used throughout this paper, evaluating on GEOM-Drugs with  $n = 200$  samples. We follow the same incremental hyperparameter search as before: sweep step size for FWDE, lock it in and additionally sweep  $\sigma$  for ALD. For PT we use the hyperparameters from Table 8.

Results are reported in Table 7. The results show that noise is critical for sampling quality, as FWDE ( $\sigma = 0$ ) achieves reasonable atom stability but near-zero molecule stability and the highest median energy, indicating that deterministic trajectories get trapped in shallow local minima. ALD

Table 6: Ablation of training objectives on GEOM-Drugs using a reduced-size model. ALD: Annealed Langevin Dynamics; PT: Parallel Tempering. No  $b(t)$  omits smoothing and static prior uses a single fixed prior during training. Best overall per metric in **bold**, best within each block underlined.

| Method                        | Sampler | Atom<br>Stability $\uparrow$ | Valid &<br>Connected $\uparrow$ | W2 Dist.<br>( $\times 10^{-2}$ ) $\downarrow$ | Diverged<br>Samples $\downarrow$ |
|-------------------------------|---------|------------------------------|---------------------------------|-----------------------------------------------|----------------------------------|
| <i>FM-based baselines</i>     |         |                              |                                 |                                               |                                  |
| Standard FM                   | FWDE    | 40.4                         | 0.00                            | 5.3                                           | 0                                |
| Standard FM                   | ALD     | 40.3                         | 0.00                            | 5.2                                           | 0                                |
| Standard FM                   | PT      | 40.1                         | 0.00                            | 5.3                                           | 0                                |
| Equivariant OT-FM             | FWDE    | 60.5                         | 1.6                             | 3.3                                           | 40                               |
| Equivariant OT-FM             | ALD     | <u>64.6</u>                  | 2.2                             | <u>0.6</u>                                    | 51                               |
| Equivariant OT-FM             | PT      | 53.7                         | <u>14.9</u>                     | 0.8                                           | 1074                             |
| <i>RFM ablations</i>          |         |                              |                                 |                                               |                                  |
| RFM, no $b(t)$ , static prior | FWDE    | 76.1                         | 9.4                             | 0.8                                           | 0                                |
| RFM, no $b(t)$ , static prior | ALD     | 79.7                         | 16.8                            | 0.6                                           | 0                                |
| RFM, no $b(t)$ , static prior | PT      | <b>85.2</b>                  | 22.4                            | <u>0.4</u>                                    | 0                                |
| RFM, static prior             | FWDE    | 78.6                         | 15.8                            | <u>0.4</u>                                    | 0                                |
| RFM, static prior             | ALD     | 79.1                         | 13.0                            | <u>0.4</u>                                    | 0                                |
| RFM, static prior             | PT      | 81.6                         | <u>22.6</u>                     | 0.5                                           | 0                                |
| RFM                           | FWDE    | 79.6                         | 23.6                            | 0.4                                           | 0                                |
| RFM                           | ALD     | 81.6                         | 33.0                            | 0.3                                           | 0                                |
| RFM                           | PT      | <u>83.6</u>                  | <b>37.7</b>                     | <b>0.2</b>                                    | 0                                |

improves substantially across all metrics, confirming that stochastic exploration helps escape these shallow basins. However, ALD tends to produce locally valid fragments rather than globally coherent molecules, as evidenced by the gap between its atom stability and Valid & Connected rate. Both PT variants close this gap, indicating that the temperature ladder successfully enables basin-to-basin transitions. A surprising observation is the significant difference between the two PT variants, as Atom Max nearly doubles molecule stability compared to Atom Avg at matched NFE (24.5% vs. 15.5% at 1080 steps). We additionally test inference-time compute scaling by increasing the NFE budget from 1080 to 1960 steps. All samplers improve, with PT + Atom Max showing the largest gains, reaching 40.0% molecule stability.

Table 7: Ablation of sampling algorithms for our RFM-trained model on GEOM-Drugs ( $n = 200$  samples). ALD: Annealed Langevin Dynamics; PT: Parallel Tempering.

| Algorithm      | (Avg.)<br>Steps | Atom<br>Stab. ( $\uparrow$ ) | Molecule<br>Stab. ( $\uparrow$ ) | Valid &<br>Connected ( $\uparrow$ ) | Diversity<br>( $\uparrow$ ) | Median<br>Energy ( $\downarrow$ ) |
|----------------|-----------------|------------------------------|----------------------------------|-------------------------------------|-----------------------------|-----------------------------------|
| FWDE           | 1080            | 88.2                         | 0.5                              | 48.0                                | 129.8                       | 21.7                              |
| FWDE           | 1960            | 88.1                         | 0.5                              | 50.0                                | 130.0                       | 20.8                              |
| ALD            | 1080            | 93.0                         | 18.0                             | 61.0                                | 110.0                       | 7.7                               |
| ALD            | 1960            | 95.5                         | 23.5                             | 70.5                                | 108.9                       | 4.6                               |
| PT + Atom Avg. | 1080            | 92.9                         | 15.5                             | 78.0                                | 116.4                       | 3.8                               |
| PT + Atom Avg. | 1960            | 95.5                         | 22.0                             | 82.5                                | 113.0                       | 1.6                               |
| PT + Atom Max  | 1080            | 95.6                         | 24.5                             | 83.0                                | 114.5                       | 2.5                               |
| PT + Atom Max  | 1960            | 97.3                         | 40.0                             | 89.5                                | 108.1                       | 1.3                               |

## E Hyperparameters & experiment Details

**Implementation.** We implement EBMol in PyTorch using PyTorch Geometric [61] for their efficient batched graph operations. SE(3) and permutation alignment uses the efficient CPU-based optimal-transport solver of [37, 36]. We use exponential moving average (EMA) for evaluation and sampling. Dataset splits for QM9 and GEOM-Drugs follow [32].

**Compute.** The QM9 model was trained for approximately 4 days on a single NVIDIA RTX 3090 Ti. The GEOM-Drugs model was trained for approximately 3 days on  $2 \times$  NVIDIA L40.

Table 8: Full hyperparameters.

| Parameter                                    | QM9                                | GEOM-Drugs        |
|----------------------------------------------|------------------------------------|-------------------|
| Network                                      |                                    |                   |
| EGNN layers                                  | 9                                  | 4                 |
| Embedding dimension                          | 256                                | 128               |
| Energy MLP layers                            | 2                                  | 2                 |
| Energy MLP hidden dim                        | 256                                | 128               |
| Total parameters                             | 5.2M                               | 2.2M              |
| Training                                     |                                    |                   |
| Optimization                                 |                                    |                   |
| Optimizer                                    | Adam                               | Adam              |
| Learning rate                                | 5e-5                               | 5e-5              |
| LR schedule                                  | Linear                             | Linear            |
| Batch size                                   | 64                                 | 16                |
| Training epochs                              | 750                                | 7                 |
| EMA decay                                    | 0.999                              | 0.999             |
| Gradient clipping                            | Yes                                | Yes               |
| RFM loss                                     |                                    |                   |
| Smoothing function $b(s)$                    | $\tanh(\gamma s)$                  | $\tanh(\gamma s)$ |
| Smoothing sharpness $\gamma$                 | 25.0                               | 25.0              |
| Energy regularization $\lambda_{\text{reg}}$ | 1e-3                               | 1e-3              |
| Prior                                        |                                    |                   |
| Position prior                               | Anisotropic Gaussian (PCA-matched) |                   |
| Atom-type prior                              | $\text{Dir}(1/K, \dots, 1/K)$      |                   |
| Sampling                                     |                                    |                   |
| Mirror-Langevin Algorithm                    |                                    |                   |
| Step size $\eta$                             | 0.05                               | 0.1               |
| Coordinate noise $\sigma_{\text{c}}$         | 0.6                                | 0.2               |
| Atom-type noise $\sigma_{\text{p}}$          | 0.6                                | 0.4               |
| Simplex floor $\epsilon$                     | 0.0015                             | 0.0005            |
| Parallel tempering                           |                                    |                   |
| Number of chains $M$                         | 8                                  | 8                 |
| Temperature levels                           | 11                                 | 11                |
| Min/Max Temperature                          | 0.05/1.0                           | 0.05/1.0          |
| Steps between swaps                          | 10                                 | 10                |
| Swaps between harvesting                     | [8, 12, 16]                        | [8, 16, 32, 64]   |
| Swap criterion                               | Atom Max                           | Atom Max          |
| Relaxation steps ( $\tau = 0$ )              | 50                                 | 200               |
| Relaxation step size                         | 0.01                               | 0.01              |

**Simplex boundary stabilization.** The Mirror Langevin Algorithm requires the mirror potential  $\Phi$  to satisfy a self-concordance-like condition (A1) of [12]: there exists  $\kappa \geq 0$  such that

$$\sqrt{2} \left\| [\nabla^2 \Phi(\mathbf{x})]^{1/2} - [\nabla^2 \Phi(\mathbf{x}')]^{1/2} \right\|_F \leq \kappa \|\nabla \Phi(\mathbf{x}) - \nabla \Phi(\mathbf{x}')\|_2. \quad (34)$$

For the negative-entropy potential  $\Phi^{\mathbf{p}}(\mathbf{p}) = \sum_i p_i \log p_i$ , the Hessian is  $\nabla^2 \Phi^{\mathbf{p}} = \text{diag}(1/p_i)$  and its square root is  $[\nabla^2 \Phi^{\mathbf{p}}]^{1/2} = \text{diag}(1/\sqrt{p_i})$ . As  $p_i \rightarrow 0$ , the left-hand side of (A1) grows as  $p_i^{-1/2}$  while the right-hand side grows as  $|\log p_i|$ , so  $\kappa = \infty$  on the full simplex interior, consistent with Table 1 of [12].

By clamping  $\mathbf{p} \geq \epsilon$  before each update, we restrict the domain to  $\Delta_{\epsilon}^{K-1} = \{\mathbf{p} \in \Delta^{K-1} : p_i \geq \epsilon \forall i\}$ , a compact subset of the simplex interior on which  $\kappa$  is finite. The choice of  $\epsilon$  involves a tradeoff: too

small and the logits  $\log p_i$  approach numerical limits, causing instability in the dual-space update; too large and the floor acts as a persistent noise source that prevents atom types from fully committing to a vertex, degrading sample quality. In effect,  $\epsilon$  controls the minimum categorical noise injected at each step via the  $1/\sqrt{p_i}$  scaling, establishing a noise floor below which the sampler cannot refine atom-type assignments. This makes  $\epsilon$  a dataset-dependent hyperparameter that requires tuning: datasets with fewer atom types tolerate smaller values, while larger type vocabularies need careful balancing to avoid both numerical instability and premature convergence. We report the values used for each benchmark in Table 8.

## F Generated samples

Figure 7 shows random samples for GEOM-Drugs using shape-steering, while Figure 8 shows unconditional samples from the same trained EBMol.

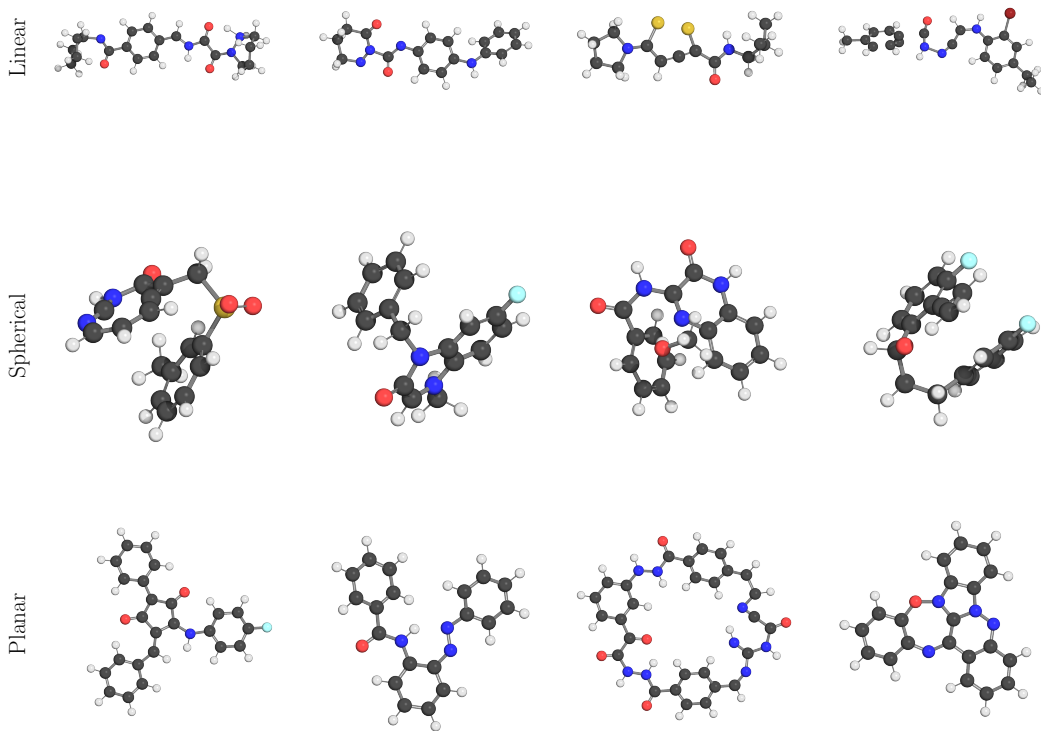


Figure 7: Random samples from an *EBMol* model trained on GEOM-Drugs using potential composition to steer the molecule shape. Samples are generated with a compute budget of 1960 NFEs.

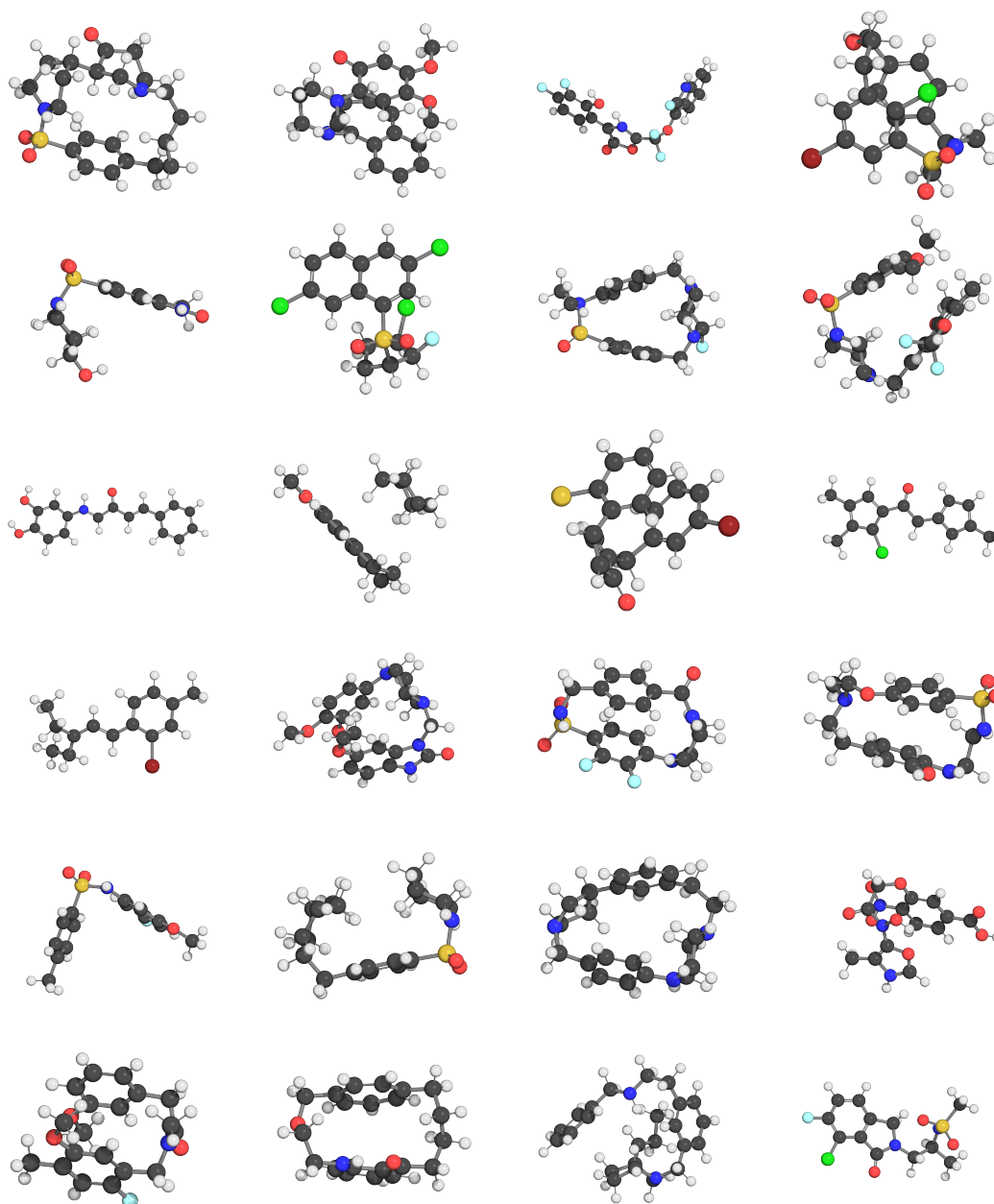


Figure 8: Random samples from an *EBMol* model trained on GEOM-Drugs. Samples are generated with a compute budget of 1960 NFEs.